

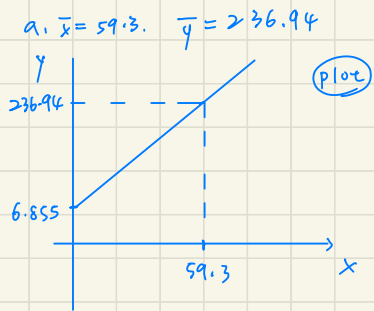
- 3.18 A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let *INCOME* be household income (thousands of dollars) and *INSURANCE* the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

小样本 t .

$$\widehat{INSURANCE} = 6.855 + 3.880 INCOME$$

(sc) (7.383) (0.112)

- Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of *INCOME* = 59.3. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.
- How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.



- Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is -0.746 .
- One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.
- Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion? $H_0: \beta = 1$

b. $a = 6.855$ $df = N - 2 = 18$
 $b = 3.88$

95% CI of β .

$$95\% = P(t_{18, 2.5\%} < \frac{b - \beta}{se(b)} < t_{18, 2.5\%})$$

$$-2.101 se(b) < b - \beta < 2.101 se(b)$$

$$-2.101 \cdot se(b) - b < -\beta < 2.101 se(b) - b$$

$$b + 2.101 se(b) > \beta > b - 2.101 se(b)$$

$$\beta \in [3.88 \pm 2.101 \cdot 0.112]$$

$$\Rightarrow \beta \in [3.6447, 4.1153]$$

In repeated sampling, there is 95% of interval cover the true value of β .

- d. Test $\beta_2 = 5$, $b \sim (\beta_2, se(b)) = (5, 0.112)$

$$H_0: \beta_2 = 5$$

$$H_1: \beta_2 \neq 5$$

$$\varphi = \frac{b - \beta_2}{se(b)} \sim t_{(18)}$$

$$RR \in [\varphi > 2.101 \vee \varphi < -2.101]$$

$$\varphi^* = \frac{3.88 - 5}{0.112} = -10 \in RR$$

Reject H_0 , No, because φ^* falls into reject area, so we shall reject H_0 and conclude $\beta \neq 5$.

c. $Cov(a, b) = -0.746$.

$y|_{x=100} = a + bx$, x 逐条

$$Var(y) = Var(a) + 100^2 \cdot Var(b) + 2 \cdot 100 \cdot Cov(a, b)$$

$$= 7.383^2 + 100^2 \cdot 0.112^2 + 200 \cdot (-0.746)$$

$$se(y) = \sqrt{7.383^2 + 100^2 \cdot 0.112^2 + 200 \cdot (-0.746)} = 5.545$$

$$\hat{y} = 6.855 + 3.88 \cdot 100 = 394.855$$

$$99\% = P(-2.878 < \frac{\hat{y} - y}{se(\hat{y})} < 2.878)$$

$$-2.878 \cdot se(\hat{y}) < \hat{y} - y < 2.878 se(\hat{y})$$

$$\hat{y} - 2.878 \cdot 5.545 < y < \hat{y} + 2.878 \cdot 5.545$$

$$y \in [394.855 \pm 2.878 \cdot 5.545]$$

$$y \in [378.8965, 410.8135] \#$$

e. $H_0: \beta_2 = 1$

$$H_1: \beta_2 \neq 1$$

$$\varphi = \frac{b - \beta_2}{se(b)} \sim t_{(18)}$$

$$RR \in [\varphi > 2.878 \vee \varphi < -2.878]$$

$$\varphi^* = \frac{3.88 - 1}{0.112} = 25.7143 \notin RR$$

Therefore, we shall reject H_0 & conclude that $\beta_2 \neq 1$

3.23 The data file *collegetown* contains data on 500 single-family houses sold in Baton Rouge, Louisiana, during 2009–2013. The data include sale price in \$1000 units, *PRICE*, and total interior area in hundreds of square feet, *SQFT*.

$n = 500$

- Using the quadratic regression model, $PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$, test the hypothesis that the marginal effect on expected house price of increasing the size of a 2000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p -value. What do you conclude?
- Using the quadratic regression model in part (a), test the hypothesis that the marginal effect on expected house price of increasing the size of a 4000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000.

Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p -value. What do you conclude?

- Using the quadratic regression model in part (a), estimate the expected price $E(PRICE|SQFT) = \alpha_1 + \alpha_2 SQFT^2$ for a house of 2000 square feet. Construct a 95% interval estimate of the expected price. Describe your interval estimate to a general audience.
- Locate houses in the sample with 2000 square feet of living area. Calculate the sample mean (average) of their selling prices. Is the sample average of the selling price for houses with $SQFT = 20$ compatible with the result in part (c)? Explain

a.

```
Residuals:
    Min       1Q   Median       3Q      Max
-383.67  -48.39   -7.50   38.75  469.70

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  93.565854    6.072226   15.41  <2e-16 ***
I(sqft^2)    0.184519    0.005256   35.11  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 92.08 on 498 degrees of freedom
Multiple R-squared:  0.7122, Adjusted R-squared:  0.7117
F-statistic: 1233 on 1 and 498 DF, p-value: < 2.2e-16
```

$$H_0: ME = 2\alpha_2 \times SQFT \leq 13 \quad SQFT = 20$$

$$H_1: ME = 2\alpha_2 \times SQFT > 13$$

$$\Rightarrow H_0: 40\alpha_2 \leq 13$$

$$H_1: 40\alpha_2 > 13$$

$$RR \in \varphi > t_{498, 5\%}$$

$$\text{邊際效應: } 7.38076 = 40\alpha_2$$

$$p\text{-value} \quad t\text{統計量} \quad -26.72875$$

$$P\text{值} \quad 1$$

$$\varphi \neq t_{498, 5\%} \text{ and } p\text{-value} > \alpha = 5\%$$

So we shouldn't reject H_0 , there is no evidence that $ME > 13$.

$$b. \quad H_0: 80\alpha_2 \leq 13; H_1: 80\alpha_2 > 13.$$

$$\varphi \neq ERR \wedge p\text{ value} < 5\%$$

$$RR \in \varphi > t_{498, 5\%} \hat{=} 1.645$$

$$\text{邊際效應: } 14.76152$$

$$t\text{統計量} \quad 4.189467$$

$$P\text{值} \quad 1.65395e-05$$

Therefore we conclude that H_0 should be rejected and ME is significantly > 13 .

c.

	fit	lwr	upr
1	167.3735	158.0481	176.6988

in repeat sampling.

There is 95% of the interval which is $[158.0481, 176.6988]$ cover the true value of $E(\text{price}|\text{sqft})$

d.

樣本中 2000 呎房子的平均售價: 163.3333

$$E(\text{price}|\text{sqft} = 20) = 163.33 \text{ falls into 95\% C.I of house price } [158.0481, 176.6988]$$

The result is compatible with the result in (c).

3.31 Data on weekly sales of a major brand of canned tuna by a supermarket chain in a large midwestern U.S. city during a mid-1990s calendar year are contained in the data file *tuna*. There are 52 observations for each of the variables. The variable *SAL1* = unit sales of brand no. 1 canned tuna, and *APR1* = price per can of brand no. 1 tuna (in dollars).

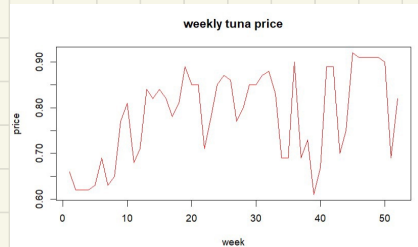
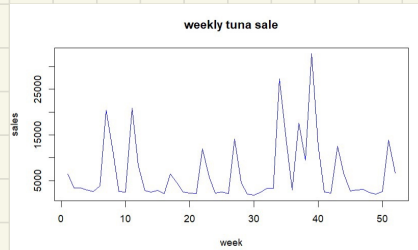
- Calculate the summary statistics for *SAL1* and *APR1*. What are the sample means, minimum and maximum values, and their standard deviations. Plot each of these variables versus *WEEK*. How much variation in sales and price is there from week to week?
- Plot the variable *SAL1* (y-axis) against *APR1* (x-axis). Is there a positive or inverse relationship? Is that what you expected, or not? Why?
- Create the variable *PRICE1* = $100APR1$. Estimate the linear regression $SAL1 = \beta_1 + \beta_2 PRICE1 + e$. What is the point estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1? What is a 95% interval estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1?
- Construct a 90% interval estimate for the expected number of cans sold in a week when the price per can is 70 cents.

$$\frac{\frac{\partial Q}{\partial P}}{\frac{Q}{P}} = \frac{\partial Q}{\partial P} \times \frac{P}{Q}$$

- Construct a 95% interval estimate of the elasticity of sales of brand no. 1 with respect to the price of brand no. 1 "at the means." Treat the sample means as constants and not random variables. Do you find the sales are fairly elastic, or fairly inelastic, with respect to price? Does this make economic sense? Why?
- Test the hypothesis that elasticity of sales of brand no. 1 with respect to the price of brand no. 1 from part (e) is minus three against the alternative that the elasticity is not minus three. Use the 10% level of significance. Clearly, state the null and alternative hypotheses in terms of the model parameters, give the rejection region, and the *p*-value for the test. What is your conclusion?

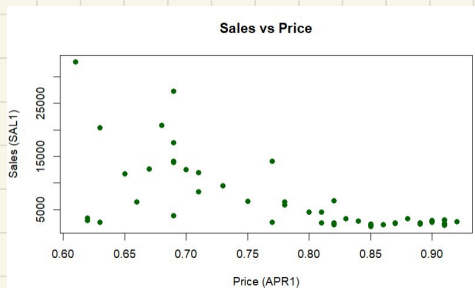
a.

	sal1	apr1
mean	6718.712	0.78250000
min	1762.000	0.61000000
max	32820.000	0.92000000
sd	6915.083	0.09803711



according to the plots on the left we can see that the variation weekly is quite large

b.



When the price of tuna is lower, the sale is high, they have inverse relation, and it also show the classic economy theory (Price effect of normal goods)

c. price = 100 APR1

```
Residuals:
    Min       1Q   Median       3Q      Max
-10869.5  -1588.3   -499.3  1626.2  18607.1

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  40714.22    6196.42   6.571 2.82e-08 ***
price1       -434.45     78.58  -5.528 1.17e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5502 on 50 degrees of freedom
Multiple R-squared:  0.3794,    Adjusted R-squared:  0.367
F-statistic: 30.56 on 1 and 50 Df,    p-value: 1.172e-06
```

$$\beta_2 \text{ 的 } t\text{-test } = b_2 = -434.45$$

95% C.I. of β_2

```
> confint(model_tuna, "price1", level = 0.95)
                2.5 %          97.5 %
price1 -592.2897 -276.6049
```

d. 90% C.I. of β_1 | $\text{Apr} = 70$

```
> new_price <- data.frame(price1 = 70)
> pred_d <- predict(model_tuna, new_price, interval = "confidence", level = 0.90)
> cat("當價格為 70 美分時，預期銷售量區間:\n")
當價格為 70 美分時，預期銷售量區間:
> print(pred_d)
      Fit      lwr      upr
1 10302.9 8624.934 11980.87
```

$$e. \text{ 3 單位 } = \frac{\partial P}{\partial Q} = \frac{\partial P}{\partial Q} \times \frac{P}{Q} = b_2 \times \frac{P}{Q}$$

$$\frac{\text{Price}}{\text{Sales}} = 78.25$$

```
> # --- (e) 價格彈性 (Elasticity) ---
> # 公式: Elasticity = b2 * (Price_mean / Sales_mean)
> b2 <- coef(model_tuna)["price1"]
> mean_p1 <- mean(tuna$price1)
> mean_s1 <- mean(tuna$all)
> elasticity <- b2 * (mean_p1 / mean_s1)
> cat("在平均價處的價格彈性為:", elasticity, "\n")
在平均價處的價格彈性為: -5.059825
```

$$\text{elasticity} = -5.0598 < -1$$

We would say that it is very elastic.
(for demand)
because tuna has plenty of substitutes.

$$f. \epsilon = \beta_2 \times \frac{P}{Q} = 0.01165 \beta_2 ; \hat{\epsilon} = 0.01165 b_2 ; \text{SE}(\hat{\epsilon}) = 0.01165 \times 78.58 \div 0.9155$$

```
> # --- (f) 假設檢定: 測試彈性是否等於 -3 ---
> # H0: b2 * (mean_p1 / mean_s1) = -3 => b2 = -3 * (mean_s1 / mean_p1)
> hyp_b2 <- -3 * (mean_s1 / mean_p1)
> se_b2 <- summary(model_tuna)$coefficients["price1", "Std. Error"]
> t_stat_f <- (b2 - hyp_b2) / se_b2
> p_val_f <- 2 * (1 - pt(abs(t_stat_f), df = df.residual(model_tuna))) # 雙尾
> cat("測試彈性是否為 -3 的 t 統計量:", t_stat_f, " P值:", p_val_f)
測試彈性是否為 -3 的 t 統計量: -2.250572 P值: 0.02884204
```

$$H_0: \epsilon = -3$$

$$H_1: \epsilon \neq -3$$

$$\varphi = \frac{\hat{\epsilon} - \epsilon}{\text{SE}(\hat{\epsilon})} \sim t_{5\%}(50)$$

$$RR \in \varphi^* > 1.645 \vee \varphi^* < -1.645$$

$$\varphi^* = \frac{0.01165(-434.45) - (-3)}{0.9155} = -2.2516 \in RR$$

We should reject H_0 , there is evidence that elasticity of tuna isn't equal to -3.